

Tata Steel Machine Failure Prediction

Project Type

Classification Machine Learning Project

Business Context & Problem Statement

In manufacturing industries, unexpected machine failures can cause production delays, increased maintenance costs, and financial losses. The objective of this project is to develop a machine learning model that can predict machine failures in advance using sensor and operational data. This enables predictive maintenance, reduces downtime, and improves operational efficiency.

Data Description

The dataset contains machine operational parameters and failure information.

Features Used

- Type
- Air Temperature [K]
- Process Temperature [K]
- Rotational Speed [rpm]
- Torque [Nm]
- Tool Wear [min]
- TWF (Tool Wear Failure)
- HDF (Heat Dissipation Failure)
- PWF (Power Failure)
- OSF (Overstrain Failure)
- RNF (Random Failure)

Target Variable

- Machine Failure (0 = No Failure, 1 = Failure)
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Exploratory Data Analysis (EDA)

Key findings from EDA:

- Machine failure cases were significantly lower than non-failure cases, indicating class imbalance.
- Temperature, rotational speed, torque, and tool wear showed strong relationships with machine failures.
- Missing values were handled appropriately.
- Outliers were analyzed and treated where necessary.
- Correlation analysis helped identify important predictive features.

(Add EDA screenshots here)

Feature Engineering & Preprocessing

- Created additional feature:
 - $\text{Temp_Difference} = \text{Process Temperature} - \text{Air Temperature}$
 - Label encoded categorical variables.
 - Applied feature scaling where required.
 - Performed train-test split using an 80:20 ratio.
 - Addressed class imbalance using SMOTE.
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Model Building Process

Three machine learning models were developed and evaluated:

Model 1: Logistic Regression

- Baseline classification model
- Applied RandomizedSearchCV for hyperparameter tuning

Model 2: Random Forest Classifier

- Ensemble learning algorithm
- Captures complex feature relationships
- Hyperparameter optimization performed

Model 3: Decision Tree Classifier

- Easy to interpret model
 - Hyperparameter tuning using RandomizedSearchCV
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Evaluation Results

Logistic Regression

- Accuracy: 83.46%
- Precision: 7.85%
- Recall: 88.37%
- F1 Score: 14.42%

Random Forest Classifier

- Accuracy: 96.84%
- Precision: 30.61%
- Recall: 79.30%
- F1 Score: 44.17%

Decision Tree Classifier

- Accuracy: 84.63%
- Precision: 7.63%
- Recall: 78.84%
- F1 Score: 13.91%

Final Model Selected

Random Forest Classifier was selected as the final model because it achieved the highest accuracy, precision, and F1 score while maintaining strong recall performance.

(Add evaluation metric charts screenshots here)

Business Impact

The developed model can:

- Predict machine failures before breakdown occurs.
 - Reduce maintenance costs.
 - Minimize production downtime.
 - Improve equipment utilization.
 - Support predictive maintenance strategies.
 - Increase overall manufacturing efficiency.
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Optional Deployment Summary

The trained Random Forest model was saved using Joblib (.pkl format). The model can be deployed using Streamlit to create an interactive web application where users can enter machine parameters and receive failure predictions in real time.

Conclusion

This project successfully developed a machine learning solution for machine failure prediction using predictive maintenance techniques. After comparing Logistic Regression, Random Forest, and Decision Tree models, Random Forest achieved the best overall performance and was selected as the final model. The solution can help manufacturing industries reduce downtime, optimize maintenance schedules, and improve operational efficiency through proactive decision-making.